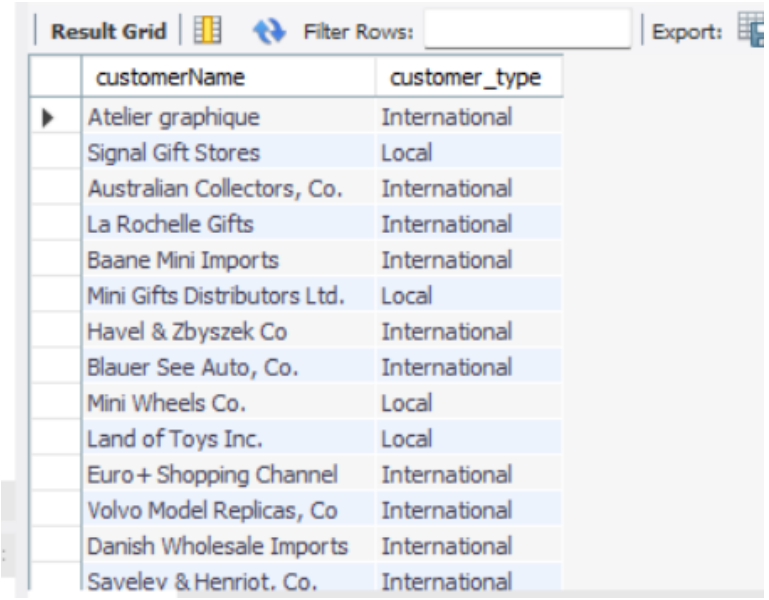
Task 2 : SQL Fundamentals

-- Mine Code

-- 1 . Transform

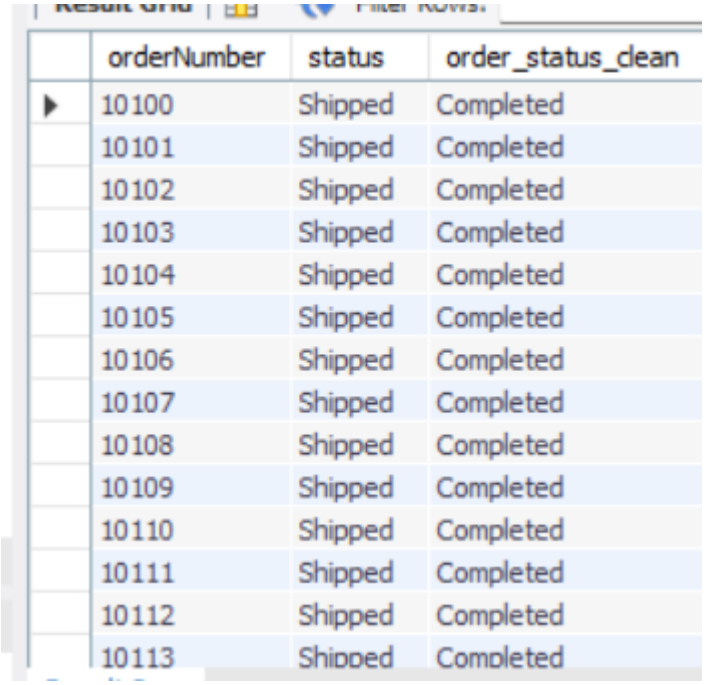
```
SELECT customerName,
CASE
   WHEN country = 'USA' THEN 'Local'
   ELSE 'International'
END AS customer_type
FROM customers;
```



Result Grid | Filter Rows: | Export:

customerName	customer_type
Atelier graphique	International
Signal Gift Stores	Local
Australian Collectors, Co.	International
La Rochelle Gifts	International
Baane Mini Imports	International
Mini Gifts Distributors Ltd.	Local
Havel & Zbyszek Co	International
Blauer See Auto, Co.	International
Mini Wheels Co.	Local
Land of Toys Inc.	Local
Euro+ Shopping Channel	International
Volvo Model Replicas, Co	International
Danish Wholesale Imports	International
Saveley & Henriot, Co.	International

```
-- order status cleaning
SELECT orderNumber, status,
CASE
   WHEN status = 'Shipped' THEN 'Completed'
   ELSE 'Pending'
END AS order_status_clean
FROM orders;
```



orderNumber	status	order_status_clean
10100	Shipped	Completed
10101	Shipped	Completed
10102	Shipped	Completed
10103	Shipped	Completed
10104	Shipped	Completed
10105	Shipped	Completed
10106	Shipped	Completed
10107	Shipped	Completed
10108	Shipped	Completed
10109	Shipped	Completed
10110	Shipped	Completed
10111	Shipped	Completed
10112	Shipped	Completed
10113	Shipped	Completed

-- 2 . Filtering

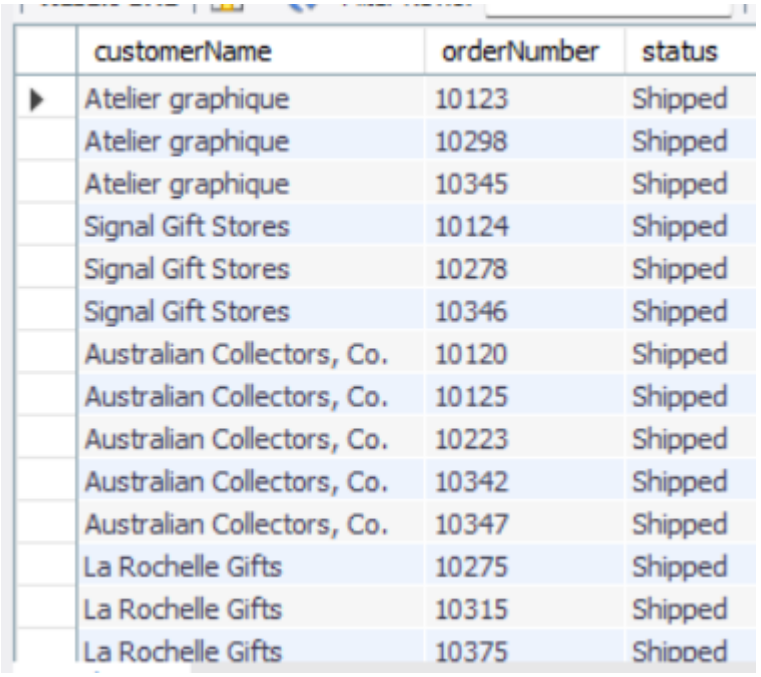
```
-- Only shipped order
SELECT *
FROM orders
WHERE status = 'Shipped';
```

orderNumber	orderDate	requiredDate	shippedDate	status	comments	customerNumber
10100	2003-01-06	2003-01-13	2003-01-10	Shipped	NULL	363
10101	2003-01-09	2003-01-18	2003-01-11	Shipped	Check on availability.	128
10102	2003-01-10	2003-01-18	2003-01-14	Shipped	NULL	181
10103	2003-01-29	2003-02-07	2003-02-02	Shipped	NULL	121
10104	2003-01-31	2003-02-09	2003-02-01	Shipped	NULL	141
10105	2003-02-11	2003-02-21	2003-02-12	Shipped	NULL	145
10106	2003-02-17	2003-02-24	2003-02-21	Shipped	NULL	278
10107	2003-02-24	2003-03-03	2003-02-26	Shipped	Difficult to negotiate with customer. We need m...	131
10108	2003-03-03	2003-03-12	2003-03-08	Shipped	NULL	385
10109	2003-03-10	2003-03-19	2003-03-11	Shipped	Customer requested that FedEx Ground is used...	486
10110	2003-03-18	2003-03-24	2003-03-20	Shipped	NULL	187
10111	2003-03-25	2003-03-31	2003-03-30	Shipped	NULL	129
10112	2003-03-24	2003-04-03	2003-03-29	Shipped	Customer requested that ad materials (such as ...	144
10113	2003-03-26	2003-04-02	2003-03-27	Shipped	NULL	124

-- 3. Joining gareko

-- Joining Customer and Orders Table

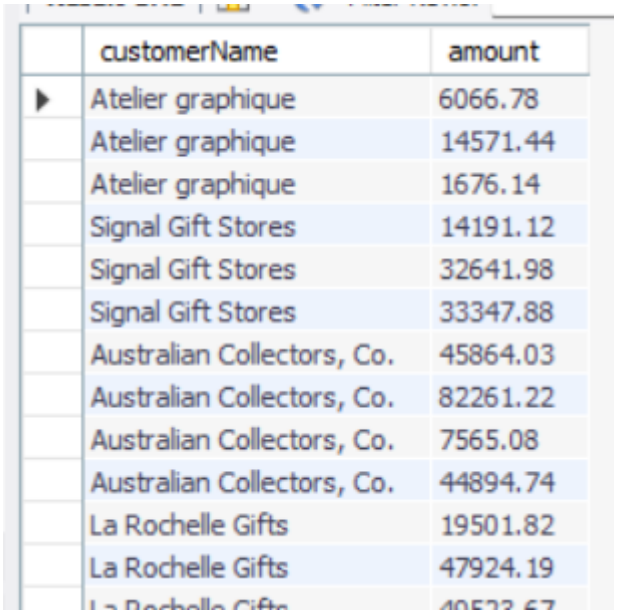
```
SELECT c.customerName, o.orderNumber, o.status
FROM customers c
JOIN orders o
ON c.customerNumber = o.customerNumber;
```



customerName	orderNumber	status
Atelier graphique	10123	Shipped
Atelier graphique	10298	Shipped
Atelier graphique	10345	Shipped
Signal Gift Stores	10124	Shipped
Signal Gift Stores	10278	Shipped
Signal Gift Stores	10346	Shipped
Australian Collectors, Co.	10120	Shipped
Australian Collectors, Co.	10125	Shipped
Australian Collectors, Co.	10223	Shipped
Australian Collectors, Co.	10342	Shipped
Australian Collectors, Co.	10347	Shipped
La Rochelle Gifts	10275	Shipped
La Rochelle Gifts	10315	Shipped
La Rochelle Gifts	10375	Shipped

-- Customers & Payments Table join gareko

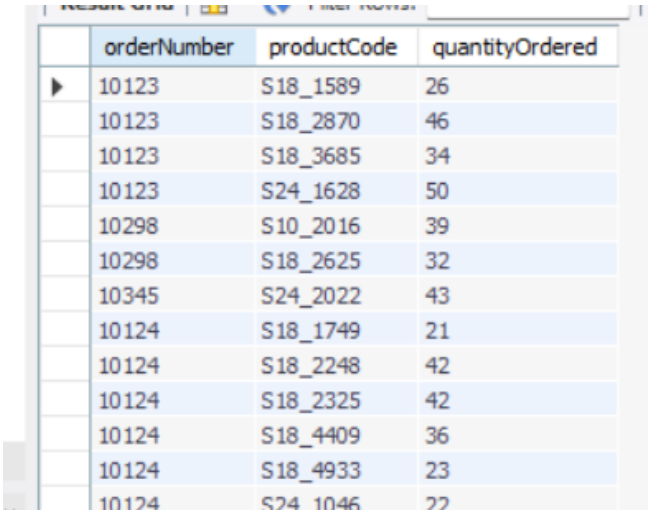
```
SELECT c.customerName, p.amount
FROM customers c
JOIN payments p
ON c.customerNumber = p.customerNumber;
```



customerName	amount
Atelier graphique	6066.78
Atelier graphique	14571.44
Atelier graphique	1676.14
Signal Gift Stores	14191.12
Signal Gift Stores	32641.98
Signal Gift Stores	33347.88
Australian Collectors, Co.	45864.03
Australian Collectors, Co.	82261.22
Australian Collectors, Co.	7565.08
Australian Collectors, Co.	44894.74
La Rochelle Gifts	19501.82
La Rochelle Gifts	47924.19

-- Customer and OrderDetails join gareko

```
SELECT o.orderNumber, od.productCode, od.quantityOrdered
FROM orders o
JOIN orderdetails od
ON o.orderNumber = od.orderNumber;
```



orderNumber	productCode	quantityOrdered
10123	S18_1589	26
10123	S18_2870	46
10123	S18_3685	34
10123	S24_1628	50
10298	S10_2016	39
10298	S18_2625	32
10345	S24_2022	43
10124	S18_1749	21
10124	S18_2248	42
10124	S18_2325	42
10124	S18_4409	36
10124	S18_4933	23

-- 4 . Aggregate

```
SELECT SUM(amount) AS total_sales
FROM payments;
```

Result Grid

total_sales
8853839.23

```
SELECT customerNumber, COUNT(orderNumber) AS total_orders
FROM orders
GROUP BY customerNumber;
```

customerNumber	total_orders
103	3
112	3
114	5
119	4
121	4
124	17
128	4
129	3
131	4
141	26
144	4
145	5

```
SELECT customerNumber, SUM(amount) AS total_payment
FROM payments
GROUP BY customerNumber;
```

Result Grid | Filter Rows:

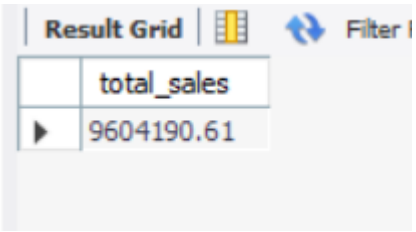
customerNumber	total_payment
103	22314.36
112	80180.98
114	180585.07
119	116949.68
121	104224.79
124	584188.24
128	75937.76
129	66710.56
131	107639.94
141	715738.98
144	43680.65
145	107446.50
146	130305.35

```
SELECT salesRepEmployeeNumber, COUNT(customerNumber) AS
total_customers
FROM customers
GROUP BY salesRepEmployeeNumber;
```



salesRepEmployeeNumber	total_customers
NULL	22
1165	6
1166	6
1188	6
1216	6
1286	7
1323	8
1337	6
1370	7
1401	10
1501	8
1504	9
1611	5
1612	5

```
SELECT SUM(quantityOrdered * priceEach) AS total_sales
FROM orderdetails;
```



Result Grid | Filter

total_sales
9604190.61

JOURNAL

In SQL, I worked with the Classic Models database. I learned how to transform, join, and aggregate data.

I used CASE statements to clean and categorize data, such as grouping customers into Local and International. I used filtering to select specific records like shipped orders.

I practiced JOIN operations to connect multiple tables such as customers, orders, and payments. This helped me understand relationships between data.

I also used aggregate functions like SUM and COUNT to calculate total sales and number of orders. I learned how to use GROUP BY to summarize data.

Challenges Faced

I faced some confusion while understanding JOIN conditions and how tables are connected. I also made small mistakes in column names and syntax, but I fixed them by checking errors and correcting them step by step.